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flux

Automatic Speech Recognition • Deepgram

`@cf/deepgram/flux`

Flux is the first conversational speech recognition model built specifically for voice agents.

> **Note**
>
> For the month of October 2025, Deepgram's Flux model will be free to use on Workers AI. Official pricing will be announced soon and charged after the promotional pricing period ends on October 31, 2025.

Model Info

Partner	Yes
Real-time	Yes

Usage

Step 1: Create a Worker that establishes a WebSocket connection

TypeScript

```
export default {
  async fetch(request, env, ctx):
Promise<Response> {
    const resp = await
env.AI.run("@cf/deepgram/flux", {
      encoding: "linear16",
      sample_rate: "16000"
    }, {
      websocket: true
    });
    return resp;
  },
} satisfies ExportedHandler<Env>;
```

Step 2: Deploy your Worker

```
npx wrangler deploy
```

Step 3: Write a client script to connect to your Worker and send audio

JavaScript

```
const ws = new WebSocket('wss://<your-
worker-url.com>');

ws.onopen = () => {
  console.log('Connected to WebSocket');

  // Generate and send random audio bytes
  // You can replace this part with a
function
  // that reads from your mic or other audio
source
  const audioData = generateRandomAudio();
  ws.send(audioData);
  console.log('Audio data sent');
};

ws.onmessage = (event) => {
  // Transcription will be received here
  // Add your custom logic to parse the data
  console.log('Received:', event.data);
};

ws.onerror = (error) => {
  console.error('WebSocket error:', error);
```

```
};

ws.onclose = () => {
  console.log('WebSocket closed');
};

// Generate random audio data (1 second of
noise at 44.1kHz, mono)
function generateRandomAudio() {
  const sampleRate = 44100;
  const duration = 1;
  const numSamples = sampleRate * duration;
  const buffer = new ArrayBuffer(numSamples
* 2);
  const view = new Int16Array(buffer);

  for (let i = 0; i < numSamples; i++) {
    view[i] = Math.floor(Math.random() *
65536 - 32768);
  }

  return buffer;
}
```

Parameters

* indicates a required field

Input

- `encoding` **string**

  Encoding of the audio stream. Currently only supports raw signed little-endian 16-bit PCM.

- `sample_rate` **string**

  Sample rate of the audio stream in Hz.

- `eager_eot_threshold` **string**

  End-of-turn confidence required to fire an eager end-of-turn event. When set, enables EagerEndOfTurn and TurnResumed events. Valid Values 0.3 - 0.9.

- `eot_threshold` **string** default 0.7

  End-of-turn confidence required to finish a turn. Valid Values 0.5 - 0.9.

- `eot_timeout_ms` **string** default 5000

A turn will be finished when this much time has passed after speech, regardless of EOT confidence.

- `keyterm` **string**

  Keyterm prompting can improve recognition of specialized terminology. Pass multiple keyterm query parameters to boost multiple keyterms.

- `mip_opt_out` **string** default false

  Opts out requests from the Deepgram Model Improvement Program. Refer to Deepgram Docs for pricing impacts before setting this to true. https://dpgr.am/deepgram-mip

- `tag` **string**

  Label your requests for the purpose of identification during usage reporting

Output

- `request_id` **string**

  The unique identifier of the request (uuid)

- `sequence_id` **integer** min 0

  Starts at 0 and increments for each message the server sends to the client.

- `event` **string**

  The type of event being reported.

- `turn_index` **integer** min 0

  The index of the current turn

- `audio_window_start` **number**

  Start time in seconds of the audio range that was transcribed

- `audio_window_end` **number**

  End time in seconds of the audio range that was transcribed

- `transcript` **string**

  Text that was said over the course of the current turn

- `words` **array**

  The words in the transcript

- `items` **object**
  - `word` **string** required

    The individual punctuated, properly-cased word from the transcript
  - `confidence` **number** required

    Confidence that this word was transcribed correctly
- `end_of_turn_confidence` **number**

  Confidence that no more speech is coming in this turn

API Schemas

The following schemas are based on JSON Schema

Input Output

```
{
    "type": "object",
    "properties": {
        "encoding": {
            "type": "string",
            "description": "Encoding of the
audio stream. Currently only supports raw
signed little-endian 16-bit PCM.",
            "enum": [
                "linear16"
            ]
        },
        "sample_rate": {
            "type": "string",
            "description": "Sample rate of the
audio stream in Hz.",
            "pattern": "^[0-9]+$"
        },
        "eager_eot_threshold": {
            "type": "string",
            "description": "End-of-turn
confidence required to fire an eager end-of-
turn event. When set, enables EagerEndOfTurn
```

```
and TurnResumed events. Valid Values 0.3 - 0.9."
        },
        "eot_threshold": {
            "type": "string",
            "description": "End-of-turn confidence required to finish a turn. Valid Values 0.5 - 0.9.",
            "default": "0.7"
        },
        "eot_timeout_ms": {
            "type": "string",
            "description": "A turn will be finished when this much time has passed after speech, regardless of EOT confidence.",
            "default": "5000",
            "pattern": "^[0-9]+$"
        },
        "keyterm": {
            "type": "string",
            "description": "Keyterm prompting can improve recognition of specialized terminology. Pass multiple keyterm query parameters to boost multiple keyterms."
        },
```

```
            "mip_opt_out": {
                "type": "string",
                "description": "Opts out requests
from the Deepgram Model Improvement Program.
Refer to Deepgram Docs for pricing impacts
before setting this to true.
https://dpgr.am/deepgram-mip",
                "enum": [
                    "true",
                    "false"
                ],
                "default": "false"
            },
            "tag": {
                "type": "string",
                "description": "Label your
requests for the purpose of identification
during usage reporting"
            },
            "required": [
                "sample_rate",
                "encoding"
            ]
        }
```

```
}
```

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